

AI Usage Journal Julian Degiorgio

Section 1: Tools declaration

Google Scholar Labs:

- Was used to help find good papers on the topic of Edge Detection for use to create the study notes.

Perplexity AI:

- The LLM was used using the Academic sources turned on to help further research on what academic papers are best suited for the task at hand.

Notebook LM:

- Was used to help verify the quality of the papers.
- Was used to create Audio overviews, Mind maps, Flash cards and Quizzes to help my personal understanding on the topic at hand.

ChatGPT:

- Was used to help structure and write sections in the study notes.
- Was used to help with ensuring clarity and readability in the created study notes.
- Was used to help structure & create content in my part in the power point presentation.
- Used to help me understand the topic and ensure that calculations in worked examples were clear.
- Help brainstorm possible questions in the quiz and help with providing good distractors from the correct answers.

Section 2: Notable prompts

Prompt 1:

I am creating the power point presentation, currently I am on the introduction.

Can you think of an example to explain edge detection with a non-technical example for the first slide of the introduction.

Summary of Prompt 1 output:

The LLM outputted the example of edge detection comparing it to looking at the outline of an object through fog.

Prompt 2:

Attached is the current version of the study notes.
Provide me with the 3x3 kernels for the Sobel edge detection operator so I can include them in my notes.

Summary of Prompt 2 output:

The LLM provided me with incorrect 3x3 kernels, the centre rows and columns of which were incorrect which was confirmed by looking through google for correct 3x3 kernels.

Prompt 3:

Create an outline for how the notes should be structured, attached are the key papers and the project brief. Remember the project brief.

Summary of Prompt 3 output:

The LLM created a bad outline which only hampered the creation of the notes when trying to create the notes in that outline.

Prompt 4:

Look at the project files.

From there, now I am working on making the quiz, remember the quiz must be fully answerable just by understanding the topic from the created notes we created.
Another member created the first 5 questions of the quiz, we must create the other 5.

I am attaching the my 5 questions I have thought up below:

6. In the Canny edge detector, what is the main purpose of double thresholding?
7. Why is Gaussian smoothing applied before the Laplacian in the Laplacian of Gaussian (LoG) method?
8. Compared with Sobel and Prewitt, what is a key advantage of the Canny detector in practical applications?
9. When using the Sobel operator, what is a primary reason it tends to be more robust to noise than the Prewitt operator?
10. Which situation is most suitable for preferring a simple gradient operator like Prewitt over Canny or LoG?

Help me think of 5 distractors for each of these questions, from which I will make use of the best 3

Remember:

Multiple-choice questions in which the incorrect answers (distractors) are based on common misconceptions or subtle edge cases.

Summary of Prompt 4 output:

The LLM outputted the 5 distractors, of which I analysed the question individually from my knowledge gained while writing the notes. The 3 distractors were then used to create the quiz.

Prompt 5:

(For Sobel Edge Detection)

I want to also include the calculation of the edge direction, if since it is $\tan^{-1}(G_y/G_x)$, in the case $G_x = 0$, what is the edge direction

Summary of Prompt 5 output:

The LLM explained that there are special cases specifically the following for the Sobel edge direction calculation using the above formula:

$$G_x = 0, G_y > 0 \Rightarrow 90^\circ$$

$$G_x = 0, G_y < 0 \Rightarrow -90^\circ \text{ or } 270^\circ$$

$$G_x = 0, G_y = 0 \Rightarrow \text{no clear edge direction}$$

Section 3: Five Questions

1. The concept I found the hardest to explain was the intuition behind edge detection without being too technical. For the introduction of the presentation, I wanted a simple example before discussing the actual operators. AI suggested explaining edge detection like looking at the outline of an object through fog. This helped because it showed edge detection is about identifying the boundaries of an object in a realistic scenario

2. AI had given me 3x3 Sobel kernels that looked correct at first, however, the central row and column were wrong as they weren't 0. Since Sobel kernels are standard and easy to accidentally copy incorrectly, this was a serious issue because it could have made the study notes technically inaccurate. I checked the kernels against external sources and corrected them before including them. This taught me not to trust mathematical outputs without verification. I then informed the model of the mistake so it can double check such cases if I ask a similar question.

3. A specific moment where I wasted my time was when I asked AI to create an outline for the study notes using the project brief and papers. When creating the notes following the outline I ended up wasting time as the provided outline did not provide notes which were easy to read. From this I learned that AI-generated structures still need strong human judgement.

4. AI genuinely helped when I was creating the quiz distractors for my questions from which I chose 3 per question. I already had the main questions outlined, but AI helped generate

several plausible incorrect answers based on misconceptions and edge cases. This made the quiz stronger because the wrong options were not random, but instead, they were mistakes a student might realistically make. The difference was that I used AI for brainstorming, then filtered the options using my own understanding from the notes and general research on the topic.

5. The question I still do not feel fully confident answering is how edge direction should be explained and implemented in all special cases, especially when $G_x = 0$ and/or $G_y = 0$ or when the gradient falls into different quadrants. I understand the basic idea, but the simple $\tan^{-1}(G_y/G_x)$ explanation hides important details.

AI Usage Journal - Daniel Briffa

Section 1: Tools Declaration

ChatGPT:

ChatGPT was used to help guide the development of my Interactive Principle Simulator for the computer vision project. It helped me plan the simulator structure, decide which features were needed, and understand how to implement Sobel, Prewitt, Laplacian or Gaussian, and Canny edge detection in a Streamlit application.

ChatGPT was also used to help debug issues during development, such as the Streamlit app showing a blank page, environment/path issues, file structure problems, and updating the app so that changes were correctly reflected when rerunning the simulator.

The tool was used to help improve the simulator by suggesting features such as image upload, built-in sample image support, parameter sliders, hover explanations, clean vs noisy comparison, method comparison tables, captions under outputs, and current settings insights.

ChatGPT was also used to help structure the walkthrough notebook. Although the walkthrough was not my main assigned deliverable, it was done to support understanding and explain the processing pipeline clearly.

For the presentation, ChatGPT was used to help summarise explanations, suggest slide layouts, and create concise text suitable for presentation format.

All AI outputs were reviewed, tested, and adapted before being used. The code was run and checked in the simulator, and the written content was edited to match the project format and my own understanding. AI was used as a support tool for guidance, explanation, debugging, and improvement, not as a direct copy-and-paste replacement for my own work.

Section 2: Notable Prompts

Prompt 1:

I have been tasked with creating the Interactive Principle Simulator for Sobel, Prewitt, Laplacian of Gaussian, and Canny. Break down my tasks into sub-tasks that I can sequentially do.

Summary of Prompt 1 Output:

The AI suggested breaking the work into stages: setting up the environment, creating the Streamlit app, implementing the edge detectors, adding parameter sliders, comparing outputs, testing failure cases, and preparing documentation. This helped me understand the full scope of my simulator task before coding.

Prompt 2:

The Streamlit app is running but the browser page is blank. What should I check?

Summary of Prompt 2 Output:

The AI helped me troubleshoot the issue by checking whether the correct file was being edited, whether Streamlit was running from the right directory, and whether the app was being rerun correctly. This helped identify that the project path and file updates needed to be checked carefully.

Prompt 3:

Can we add hover explanations beside each slider so users understand what each parameter does?

Summary of Prompt 3 Output:

The AI suggested using Streamlit help text for each slider. This improved the usability of the simulator because users can understand what blur, noise, Canny thresholds, and LoG threshold do without needing separate instructions.

Section 3: Five Questions

1. The concept I found hardest to turn into an interactive feature was parameter sensitivity. I understood that changing thresholds affects edge detection, but I needed help deciding how to show this clearly in the simulator. AI suggested using sliders and a clean vs noisy comparison, which helped make the effect visible rather than only theoretical.

2. A useful moment was when AI helped me debug the blank Streamlit page. At first, the app was running but nothing appeared in the browser. The AI guided me to check whether I was editing the correct `app.py` file and running Streamlit from the correct project folder. This helped me understand that coding issues are not always caused by the code itself, but sometimes by file paths, environments, or execution commands.

3. AI helped improve the simulator interface by suggesting captions, tooltips, a comparison table, and current settings insight. These suggestions made the simulator more understandable for users because it no longer just displayed outputs, but also explained what the viewer should notice. I reviewed these suggestions and only kept the ones that made the simulator clearer.

4. For the Canny presentation slides, AI helped me summarise the pipeline into simple slide-friendly wording. I did not copy the explanations directly without checking them. I compared the output with the project material and adjusted the wording so it matched the visual slides and remained easy for the audience to understand.

5. The area I still feel I could improve is explaining the deeper mathematical details behind the Canny gradient calculation and non-maximum suppression. I understand the overall pipeline and how to demonstrate it visually, but the exact pixel-level calculations are more complex. AI helped me simplify this for presentation purposes, but I would still need further study to explain every mathematical step in full detail.

AI Usage Journal - Samuel Bugeja

Section 1: Tools Declared

ChatGPT (GPT-5)

- Used to help extend the study notes, improve explanations, generate worked examples, refine comparisons between edge detectors, and assist with quiz question generation.

Tool: NotebookLM

- Used to summarise uploaded papers and extract general information about Sobel, Prewitt, LoG, and Canny operators. Also used to brainstorm quiz ideas and misconceptions.

Section 2: Notable Prompts

Prompt 1:

"i want to continue where my friend left off on the study notes, how can i continue? give me a ideas which i can add"

Summary:

The AI suggested adding comparative analysis, parameter sensitivity, failure cases, and evaluation sections. These additions helped transform the notes from mostly descriptive

theory into a more analytical learning resource.

Prompt 2:

“can you provide a worked example for gaussian aswell”

Summary:

The AI generated a step-by-step Gaussian smoothing example using a 3×3 kernel and numerical calculations. This became part of the mathematical and computational section of the notes and helped satisfy the requirement for worked numerical examples.

Prompt 3:

“can you improve the quick reference section?”

Summary:

The AI generated a concise comparison table summarising Sobel, Prewitt, LoG, Canny, and Gaussian filtering. This improved the usability of the notes by creating a revision-oriented cheat sheet instead of only long-form explanations.

Prompt 4:

“what else can i possibly add”

Summary:

The AI identified missing requirements from the brief, including analogies, real-world examples, and worked examples. This changed the direction of the document because it highlighted that the notes were too theory-heavy and lacked application-focused explanations.

Prompt 5 (NotebookLM):

“Summarise the differences between Sobel, Prewitt, Laplacian of Gaussian, and Canny edge detectors. Their strengths, weaknesses, and common misconceptions.”

Summary:

NotebookLM produced concise summaries from uploaded papers and highlighted common misconceptions such as confusing gradient direction with edge direction and misunderstanding why LoG detects zero-crossings instead of gradient maxima. These insights influenced both the study notes and several quiz distractors.

Section 3: 5 Questions

1. The hardest concept to explain was the Laplacian of Gaussian and the idea of zero-crossings. Sobel and Prewitt are easier because people intuitively understand “strong brightness change = edge.” LoG was harder because it detects where the second derivative changes sign rather than where intensity simply changes most. Explaining why Gaussian smoothing is necessary before applying the Laplacian was also difficult, especially because second derivatives naturally amplify noise.

2. One issue occurred when AI generated explanations for gradient direction in the Sobel

worked example. The wording initially confused edge direction with gradient direction, which are perpendicular rather than identical. Another issue was overly simplified explanations of LoG that ignored zero-crossings entirely. To fix this, I cross-checked the information against the research papers used in the notes and rewrote the explanations manually.

3. A significant waste of time happened when trying to generate quiz questions entirely through AI in one go. Many generated questions were too obvious, repetitive, or tested memorisation instead of understanding. Some distractors were unrealistic and easy to eliminate immediately. I ended up rewriting many of the questions manually and only using

AI for brainstorming misconceptions and edge cases.

4. AI was most useful when expanding and restructuring the study notes. After the theoretical sections were already written, AI quickly suggested missing areas such as failure cases, comparative analysis, implementation notes, analogies, and quick-reference summaries. This improved the quality and completeness of the notes much faster than manually planning the structure ourselves.

5. One area we still do not feel completely confident about is how to formally evaluate edge detector quality objectively across different images. We understand the general strengths and weaknesses of Sobel, Prewitt, LoG, and Canny, but choosing the “best” detector depends heavily on image conditions, noise levels, and application context.